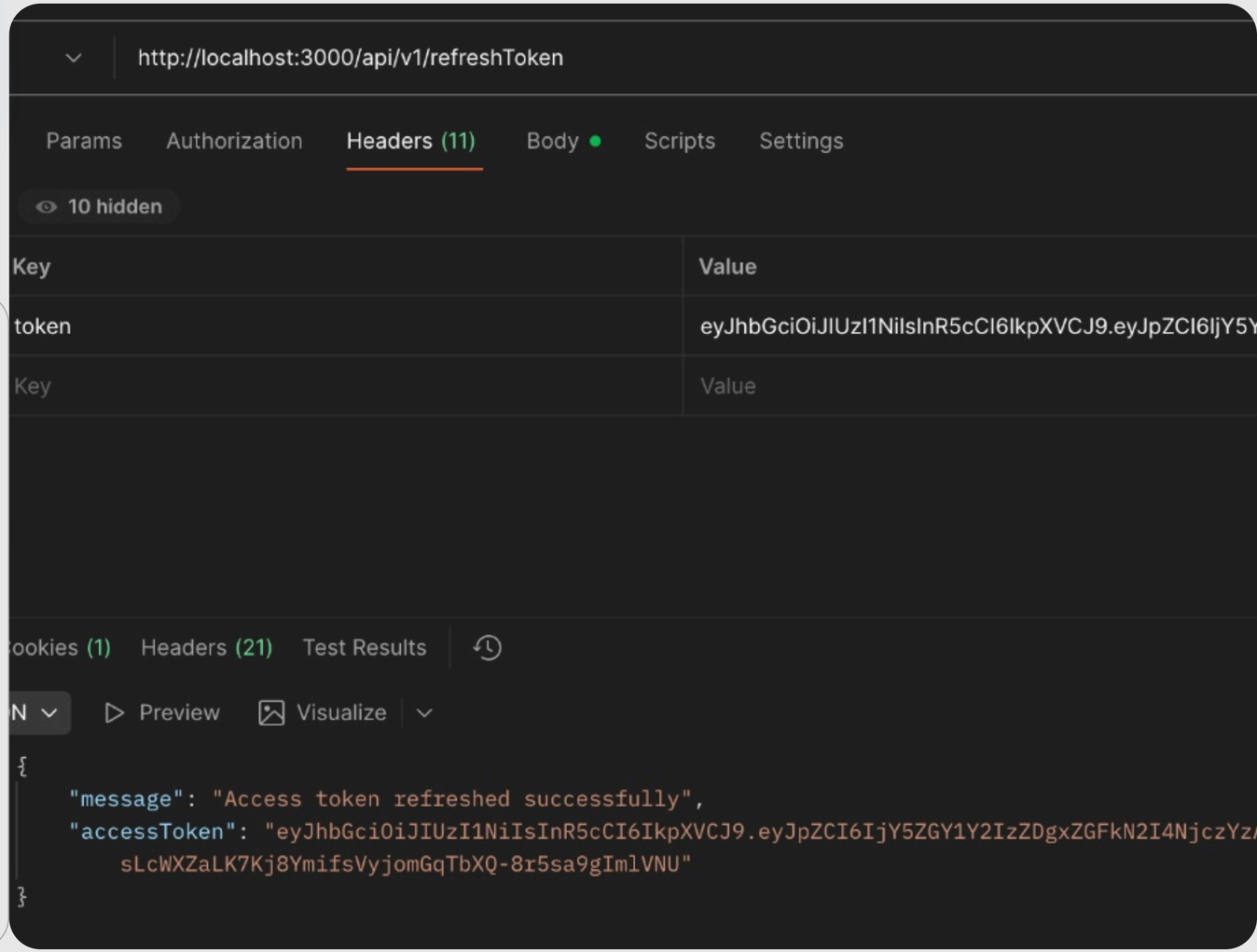
CYBER SECURITY

TOKEN-BASED AUTHENTICATION SYSTEM

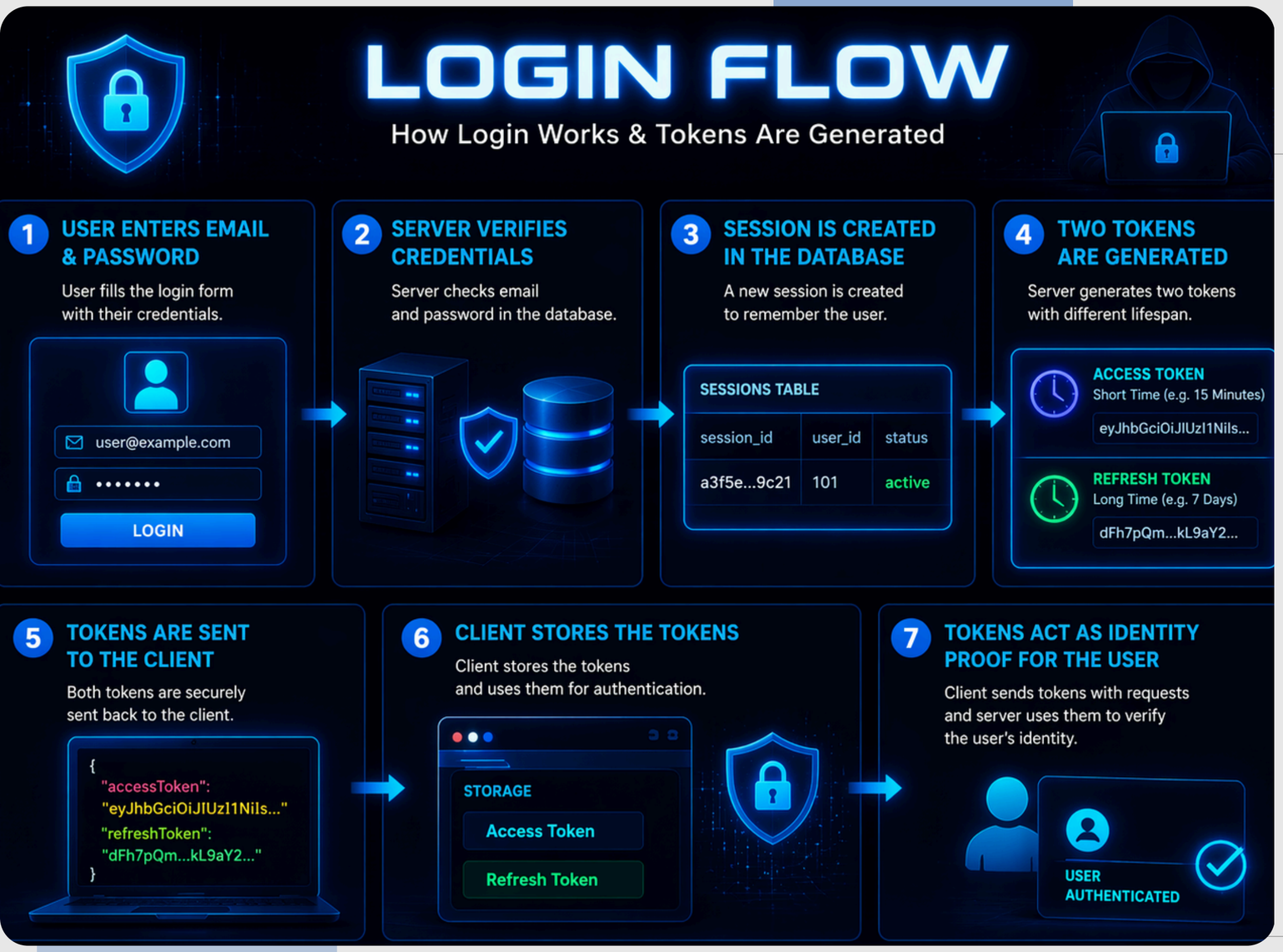
Presented By:
Dhruv Singh
Harsh Chhatri
Khushi Chadda
Lakshya Maheshwari

UNDERSTANDING CYBER SECURITY

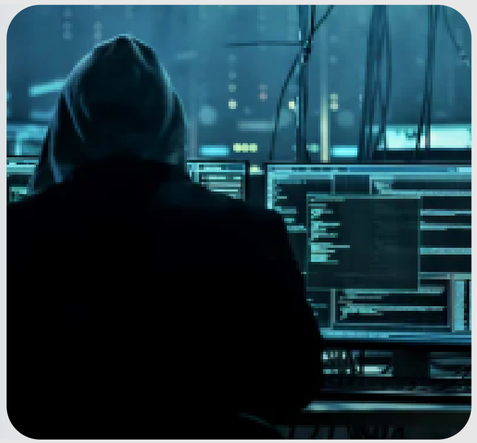
- This project demonstrates how login systems work using tokens
- It explains both secure flow and attack scenarios
- Focus is on Access Token and Refresh Token mechanism
- Shows how attackers can misuse stolen tokens
- Also implements security solutions like IP binding
- Built as a web-based application for learning cybersecurity



LOGIN FLOW



- User enters email and password to login
- Server verifies the credentials
- A session is created in the database
- Two tokens are generated:
- Access Token (short time)
- Refresh Token (long time)
- Tokens are sent to the client
- These tokens act like identity proof for the user

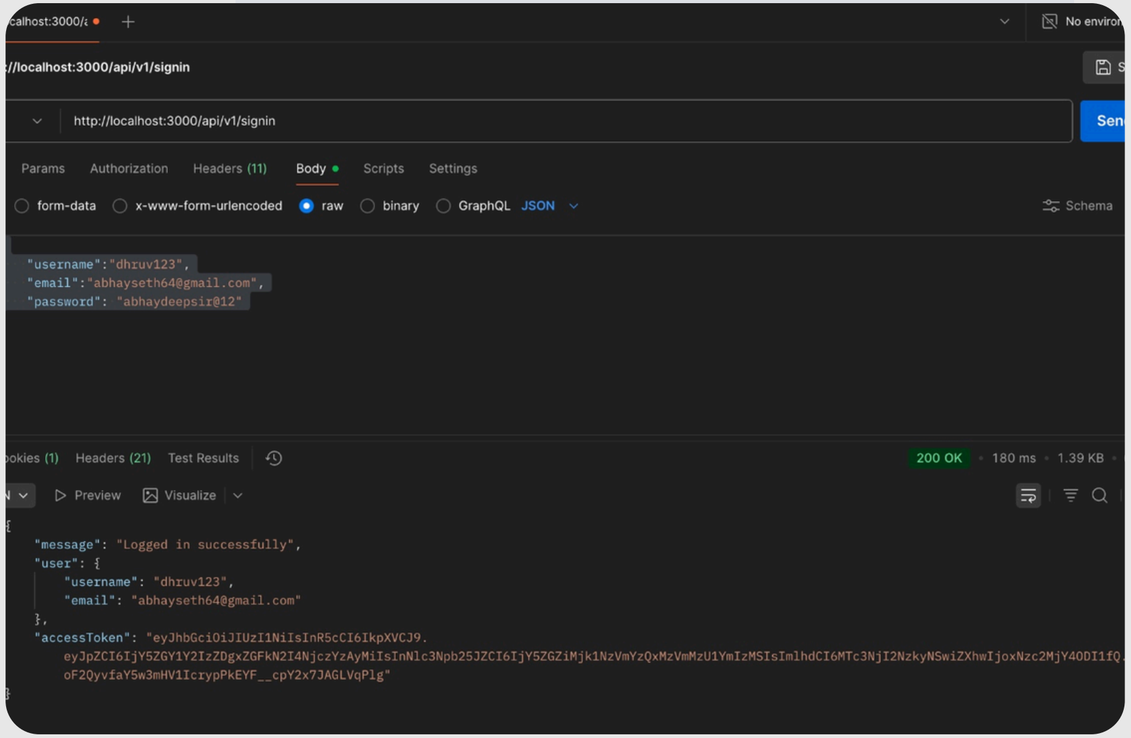


API REQUEST FLOW

- User sends request to server using Access Token
- Server verifies the Access Token
- If valid → user gets requested data
- If invalid → request is rejected
- Access Token is used for every action
- Keeps system fast and secure

TOKEN EXPIRY & REFRESH FLOW

- Access Token expires after short time
- User sends Refresh Token to server
- Server verifies refresh token
- Checks session stored in database
- Generates new Access Token
- Also rotates Refresh Token for security



ATTACK SCENARIO

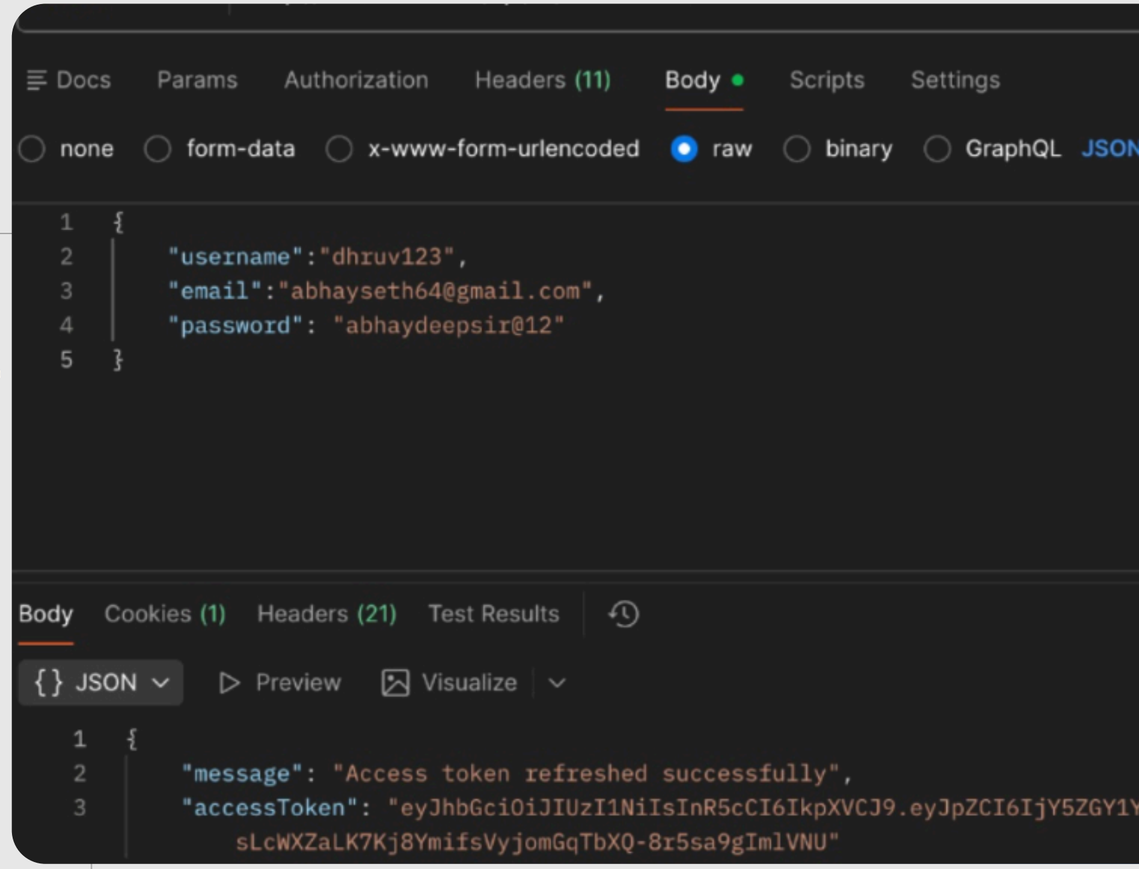
If attacker steals Access Token

They can access user data

If attacker steals Refresh Token

They can generate new tokens

- This allows attacker to stay logged in
- This is called session hijacking
- It is a major security risk in web apps
- Tokens alone are not fully safe



SECURITY SOLUTION





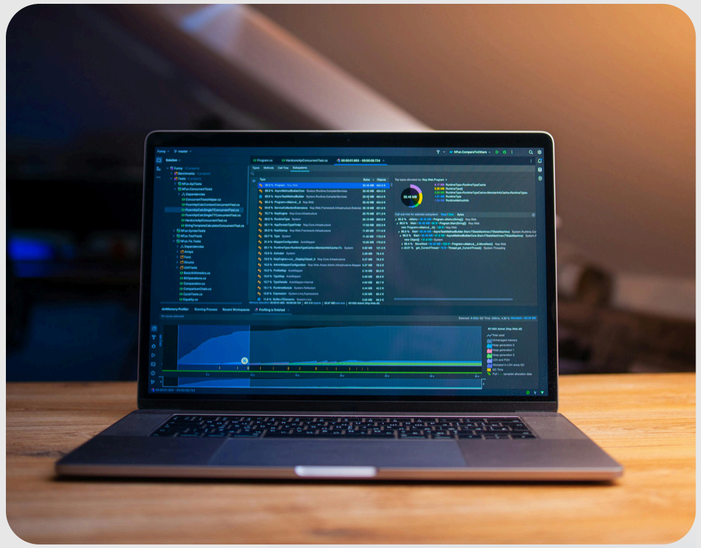
- Tokens are linked with user's IP address
- Server stores IP during login
- Every request checks the IP address
- If IP changes → request is blocked
- Prevents misuse of stolen tokens
- Adds an extra layer of security

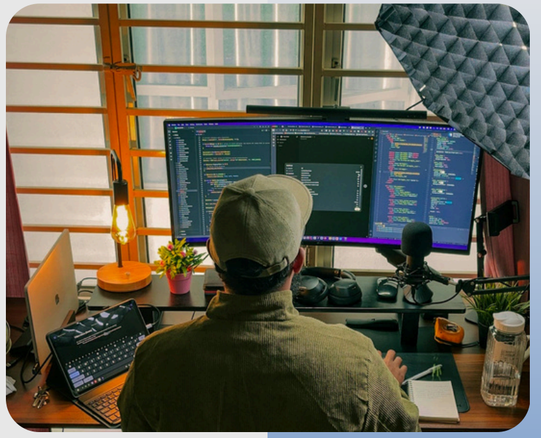


ADDITIONAL SECURITY FEATURES

- Refresh Token Rotation (new token every time)
- HTTP-only cookies for secure storage
- Session stored in database for validation
- Interceptor handles token expiry automatically
- Logout invalidates all tokens
- Improves both security and user experience







CONCLUSION

- Token-based authentication is widely used
- It improves performance and scalability
- But has security risks if not handled properly
- Our system demonstrates both attack and solution
- IP binding and token rotation improve security
- Helps understand real-world cybersecurity concepts